



UNIVERSITI
MALAYA

SOLAR-HEAT-WATER HARVESTER

✓ KEY INFORMATION

Patent Number: UI 2023002890

Technology Readiness Level: TRL 3

Market Readiness Assessment: MR 3

Location: Universiti Malaya, Malaysia



✓ TECHNOLOGY OVERVIEW

This invention introduces a **low-cost, easy-to-assemble solar-heat-water harvester** designed to **capture both solar energy and rainwater**. The pyramid-shaped structure uses **transparent plastic (perspex)** and **solar panels** to convert sunlight into electricity while collecting rainwater through an efficient drainage system. The system also uses heat convection to power a fan, generating additional electricity using warm air currents. It is ideal for promoting **green energy and water sustainability**.

✓ FEATURES AND SPECIFICATIONS

Pyramid Structure: Maximizes sunlight capture and natural airflow

Photovoltaic Solar Panels: Converts sunlight into electricity

Heat Convection Fan: Driven by rising hot air or lateral winds

Rainwater Collection: Channels water to edge drains via aluminum gutters

Materials:

- Transparent perspex for sunlight penetration
- Aluminum collector – rustproof, lightweight, and durable

Scalable Design: Can be resized for homes, buildings, or public spaces

Portable & Low Maintenance: Easy to install, durable under various climates

✓ POTENTIAL APPLICATION

Rooftop Installations on homes, schools, or factories

Green Infrastructure Projects in cities or farms

Emergency Shelters for off-grid energy and water

Public Parks or Community Spaces for water and energy conservation

Sustainable Housing Developments and eco-villages

✓ UNIQUE VALUE PROPOSITIONS

Dual Functionality: Generates electricity and collects rainwater

Low-Cost and Easy Assembly: Suitable for rural and urban deployment

Environmentally Friendly: Supports sustainability with zero emissions

Scalable and Modular: Adaptable to small or large installations

Off-Grid Potential: Ideal for disaster zones or remote communities

YOU HAVE INQUIRY? >>>

For more information, contact:
Dr. Lee Ching Shya (RTTP) | leecs@um.edu.my
Dr. Nurul Fauzani Jamaluddin | nfauzani@um.edu.my
UM Centre of Innovation and Enterprise (UMCIE)
umcie@um.edu.my
Tel: +603-79677351